

Problem 274: Delayed Graphics

Difficulty: Easy

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Problem Background

Your computer monitor may appear to display information and graphics instantly in response to your actions, but in reality there's a slight delay. On a laptop computer, this delay is usually negligible, since the laptop's processor runs very quickly and there's often no reliance on other systems. However, on an aircraft, a processor may not run as quickly in order to save power, and there may be delays as the onboard computer waits for information from its sensors and ground traffic control. Since these delays can be noticeable to the pilots (and can be a serious problem in emergencies), aircraft manufacturers like Lockheed Martin may have requirements to provide information to the pilot within a certain amount of time from when an input is received.

To meet these requirements, we can run tests to see how fast the code is currently running, compared to the maximum allowed time. If the current response time is too slow, developers can use software profilers to identify and resolve bottlenecks that slow the software down.

Problem Description

You will be provided with the results of a performance test on the guidance systems for a new fighter jet built by Lockheed Martin Skunk Works. These tests checked how long it took for an input from the pilot to be received by the guidance computer, and then how long it took for the guidance computer to return a response to the pilot's displays. Using that information, you'll need to determine if the response was quick enough to meet the given performance requirement. If the delay was longer than the maximum specified by the requirement, you'll need to calculate the difference between those times to give developers an idea of how much improvement is needed.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include two lines, as follows:

- The first line will contain a positive integer, representing the maximum delay time permitted (in milliseconds) according to the software requirements
- The second line will contain two positive integers, separated by a space, respectively representing the input delay time and the processing delay time.

```
4
100
50 75
120
100 100
80
30 35
90
45 45
```

Sample Output

For each test case, your program must print a single line, as follows:

- If the provided delay times are less than or equal to the specified maximum delay, print the string "PASS"
- Otherwise, print the difference (in milliseconds) between the actual delay time and the maximum allowed delay time

```
25
80
PASS
PASS
```